

C# BACKEND TRACK | OOP PHASE

Task 1

System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

System Generated	System Generated	The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.
User Input	User Input	The user types this value directly at the keyboard when prompted by the system.
Default Value	Default Value	A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".
Calculated	Calculated	Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.
From List	From List	The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.

Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently **active loans**, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: <u>Book</u>		
Property Name	Data Type	Source Tag
<u>Title</u>	<u>string</u>	<u>User input</u>
<u>Author</u>	<u>string</u>	<u>User input</u>
<u>genre</u>	<u>string</u>	<u>User input</u>
<u>fix_num_ofcopies</u>	<u>int</u>	<u>Default Value</u>
<u>book_id</u>	<u>int</u>	<u>sys generate</u>
<u>Total_borrow_count</u>	<u>int</u>	<u>Calculated</u>
<u>isAvailable</u>	<u>bool</u>	<u>Default Value</u>

Entity Name: <u>Member</u>		
Property Name	Data Type	Source Tag
<u>Member_id</u>	<u>int</u>	<u>Sys generate</u>
<u>Name</u>	<u>string</u>	<u>User input</u>
<u>Register_date</u>	<u>string</u>	<u>sys generate</u>
<u>Phone_num</u>	<u>int</u>	<u>User input</u>
<u>Email</u>	<u>string</u>	<u>User input</u>
<u>isActive_loans</u>	<u>bool</u>	<u>default value</u>

Entity Name: <u>loan</u>		
Property Name	Data Type	Source Tag
<u>loan_id</u>	<u>int</u>	<u>Sys generate</u>
<u>Member_id</u>	<u>int</u>	<u>from list</u>
<u>Book_id</u>	<u>int</u>	<u>from list</u>
<u>BorrowDate</u>	<u>string</u>	<u>sys generate</u>
<u>DueDate</u>	<u>string</u>	<u>default value</u>
<u>ReturnDate</u>	<u>string</u>	<u>sys generate</u>
<u>LoanStatus</u>	<u>string</u>	<u>default value</u>

Entity Name: <u>Librarian</u>		
Property Name	Data Type	Source Tag
<u>LibrarianID</u>	<u>int</u>	<u>Sys generate</u>
<u>Name</u>	<u>String</u>	<u>User input</u>
<u>Email</u>	<u>String</u>	<u>User input</u>

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

1 Member : can have many Loans.
 1 Book : can appear in many Loans.
 1 Loan : belongs to 1 Member and 1 Book.
 1 Loan: may have 1 Fine.
 1 Member : can make many Reservations.
 1 Book : can have many Reservations.
 1 Reservation : belongs to 1 Member and one Book.

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	RegisterMember	Registers new member
2	AddBook	Adds new book
3	BorrowBook	Borrows a book.
4	ReturnBook	Returns a book.
5	ReserveBook	Reserves a book..
6	ViewActiveLoans	Shows active loans.
7	ViewOverdueLoans	Shows overdue loans.
8	MostBorrowedBooks	Shows most-borrowed books.
9	CheckBookAvailability	Checks if a book is available.
10	UpdateBorrowCount	total borrow count

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest		
Property Name	Data Type	Source Tag (fill in)
guestId	int	sys generate
guestName	string	User input
guestPhone	string	User input
guestEmail	string	User input
guestNationality	string	user input
registrationDate	string	sys generate

Room		
Property Name	Data Type	Source Tag (fill in)
roomId	int	<u> Sys generate </u>
roomNumber	string	<u> user input </u>
roomType	string	<u> From list </u>
floorNumber	int	<u> User input </u>
nightlyPrice	decimal	<u> Calculated </u>
isAvailable	bool	<u> default value </u>

Booking		
Property Name	Data Type	Source Tag (fill in)
bookingId	int	<u> sys generate </u>
guestId	int	<u> from list </u>
roomId	int	<u> from list </u>
checkInDate	string	<u> User input </u>
checkOutDate	string	<u> User input </u>
numberOfNights	int	<u> Calculated </u>
totalPrice	decimal	<u> calculated </u>
status	string	<u> deflate value </u>

CheckInRecord		
Property Name	Data Type	Source Tag (fill in)
recordId	int	sys generate
bookingId	int	from list
guestId	int	from list
roomId	int	from list
actualCheckIn	string	default value
actualCheckOut	string	user input
finalAmount	decimal	Caculated

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

- Write down the relationships
- 1 Guest : can have many Bookings.
 - 1 Room : can have many Bookings over time / but only 1 active booking at time.
 - 1 Booking : belongs to 1 Guest and 1 Room.
 - 1 Booking : can create 1 CheckInRecord .
 - 1 CheckInRecord :for 1 Booking, 1 Guest, 1 Room.

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	RegisterGuest	Adds a new guest.
2	AddRoom	Adds a new room
3	ViewAvailableRooms(	Shows available rooms.
4	BookRoom	Creates a room booking
5	CalculateNumOfNights	Calculates the stay duration.
6	CalculateTotalPrice	Calculates the total cost
7	CheckInGuest	Records guest check-in
	CheckOutGuest	Records guest check-out
	CancelBooking	Cancels a booking

#	Service Name	What it does
8	<u>RoomAvailability</u>	<u>Updates room availability status.</u>
9	<u></u>	<u></u>